



Lesson 7.3 — Sine and Cosine Functions

Big idea: Sine and cosine are the y- and x-coordinates of the unit circle as the angle θ changes. As functions of θ , they trace out smooth waves.

Quick review

Definition: for angle θ , $\sin \theta$ is the y-coordinate and $\cos \theta$ is the x-coordinate on the unit circle.

Domain: all real numbers (θ unbounded). **Range:** $[-1, 1]$ for both.

Period: both have period 2π (or 360°). After that they repeat.

Even/odd: cosine is even ($\cos(-\theta) = \cos \theta$), sine is odd ($\sin(-\theta) = -\sin \theta$).

Pythagorean identity: $\sin^2 \theta + \cos^2 \theta = 1$.

Worked example

Worked example. Find $\sin(7\pi/6)$.

$7\pi/6$ is in Q3, reference angle $\pi/6$. In Q3 sine is negative $\Rightarrow \sin(7\pi/6) = -\sin(\pi/6) = -1/2$.

Warm-ups

1. Find $\sin(\pi/2)$.
2. Find $\cos(\pi)$.
3. Find $\cos(0)$.

Practice

1. Find $\sin(2\pi/3)$.
2. Find $\cos(5\pi/3)$.
3. Use the Pythagorean identity: if $\sin \theta = 3/5$ and θ is in Q2, find $\cos \theta$.

Challenge

1. Show that $\cos(-\theta) = \cos(\theta)$ using the unit circle.
2. Solve $\sin \theta = 1/2$ for $0 \leq \theta < 2\pi$.



Answer key — Lesson 7.3

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

- 1.
- 1.
- 1.

Practice

- Q2: positive sine. $\sin(2\pi/3) = \sin(\pi/3) = \sqrt{3}/2$.
- Q4: positive cosine. $\cos(5\pi/3) = \cos(\pi/3) = 1/2$.
- $\cos^2\theta = 1 - 9/25 = 16/25 \Rightarrow \cos\theta = \pm 4/5$. Q2: cosine negative $\Rightarrow \cos\theta = -4/5$.

Challenge

- Reflecting the unit circle point $(\cos\theta, \sin\theta)$ across the x-axis (which is what $\theta \rightarrow -\theta$ does) gives $(\cos\theta, -\sin\theta)$. The x-coordinate is unchanged, so $\cos(-\theta) = \cos\theta$.
- $\theta = \pi/6$ (Q1) and $5\pi/6$ (Q2).